

PHISHING EMAIL DETECTION & AWARENESS REPORT

Task2

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TABLE OF CONTENTS

INTRODUCTION	2
OBJECTIVES.....	3
TOOLS USED	3
PHISHING EMAIL SAMPLE ANALYSIS.....	4
EXAMPLE ANALYSIS WRITE-UP:	4
EMAIL HEADER ANALYSIS.....	4
RISK CLASSIFICATION TABLE	4
HOW PHISHING WORKS.....	4
PREVENTION & AWARENESS GUIDLINES.....	4
EMPLOYEES DO'S AND DON'TS.....	5
CONCLUSION:	5

INTRODUCTION

Phishing is a cyber-attack where attackers impersonate trusted organizations to trick users into revealing sensitive information such as passwords, banking details, or OTPs

The purpose of this report is to analyze phishing email samples, identify common phishing indicators, classify email risks, and provide awareness guidelines to help users recognize and avoid phishing.

OBJECTIVES

- Analyze phishing email samples
- Identify phishing indicators
- Investigate suspicious sender domains and links
- Classify email risks
- Provide phishing prevention awareness
- Educate users on safe email practices

TOOLS USED

- toolbox.googleapp.com
- GitHub Phishing Repositories

PAYPAL PHISHING EMAIL SAMPLE ANALYSIS

SAMPLE 1: PAYPAL PHISHING EMAIL

Email details:

- Sender: security-alert@paypal-support.com
- Subject: Urgent: Suspicious Activity Detected on your PayPal Account
- Recipient: youremail@example.com

PHISHING INDICATORS IDENTIFIED

- Suspicious sender domain:

The sender uses paypal-support.com instead of the official paypal.com domain

- Urgent language:

The email creates pressure by demanding immediate action within 24 hours.

- Fear tactics:

The email threatens permanent account suspension.

- Generic greeting:

Use of "Dear Customer" instead of the recipient's real name.

- Request for verification:

Attempts to trick users into providing account information

- Lack of verification contact:

No official support contact information provided.

RISK CLASSIFIACATION

CATERGORY	RESULT
Email type	Phishing email
Risk Level	High
Threat Type	Credential theft

HOW THE ATTACKS WORKS:

This phishing email impersonates PayPal and attempts to create fear and urgency. Victims may click on a malicious verification link and unknowingly provide their login credentials to attackers.

Awareness Tips:

- Verify sender email domains carefully
- Never click suspicious links
- Visit official websites directly
- Be cautious of urgent threats
- Enable multi-factor authentication

PAYPAL E-mail Header Analysis

Email header analysis helps investigators determine whether e-mail is legitimate. Suspicious phishing emails often fail authentication checks, such as SPF, DKIM and DMARC

Security check.	Result.
SPF.	Failed.
DKIM.	Failed.
DMARC.	Failed.

Explanation.

- SPF failure indicates that the sender is not authorized to send emails on behalf.
- DK IM failure suggests that the e-mail signature could not be verified.
- DMA RC failure indicates that the e-mail failed authentication policies

These failures are common indicators of phishing and e-mail spoofing attacks.

The PayPal phishing e-mail phishing was analyzed using the Google Admin Toolbox Message Header analyzer. The results showed SPF, DKIM, and DMARC authentication failures, which indicate that the sender was not authorized to send emails on behalf of the domain.

The sender domain “paypal-support.com” is not an official PayPal domain and has failed multiple email authentication checks. These failures are common indicators of phishing and email spoofing attacks.

MessageId	001a113000b5117a@paypal-support.com
Created at:	5/27/2025, 11:41:59 AM GMT+2 (Delivered after 1 sec)
From:	PayPal Security <security-alert@paypal-support.com>
To:	youremail@example.com
Subject:	Urgent: Suspicious Activity Detected on Your PayPal Account
SPF:	fail with IP Unknown! Learn more
DKIM:	fail with domain paypal-support.com; Learn more
DMARC:	fail Learn more

#	Delay	From *	To *	Protocol	Time received
0	1 sec	unknown	→ mail.example.com	SMTP	5/27/2025, 11:42:00 AM GMT+2

PAYPAL PHISHING EMAIL SAMPLE ANALYSIS

SAMPLE 2: FNB BANKING PHISHING

Email details:

- Sender: support@fnb-securelogin.com
- Subject: Important Notice: Your Bank Account Has Been Locked
- Recipient: youremail@example.com

PHISHING INDICATORS IDENTIFIED:

- Fake sender domain:

The sender domain does not belong to the Official Bank.

- Suspisoius URL

The link redirects users to a fake banking website.

- Urgent language

the email pressures the victim to act immediately

- Fear tactics

Threatens permanent disabling of Banking services.

- Generic greeting identity verification request:

Attempts to steal banking credentials.

Risk classification.

CATEGORY	RESULT
Email Type	Phishing email
Risk Level	High
Threat type	Banking Fraud

HOW THE ATTACK WORKS

The attacker impersonates a legitimate bank and tricks users into clicking on a malicious link to verify the message. Victims may unknowingly provide banking logging initial credentials and personal information.

Awareness Tips

- Enable multi-factor authentication
- Visit official website directly
- Verify sender email domain carefully

- Never click suspicious links

FNB E-mail Header Analysis

Email header analysis helps investigators determine whether e-mail is legitimate. Suspicious phishing emails often fail authentication checks, such as SPF, DKIM, DMARC

Security check.	Result.
SPF.	Failed.
DKIM.	Failed.
DMARC.	Failed.

Explanation.

- SPF failure indicates that the sender is not authorized to send emails on behalf.
- DKIM failure suggests that the e-mail signature could not be verified.
- DMARC failure indicates that the e-mail failed authentication policies

These failures are common indicators of phishing and e-mail spoofing attacks.

The FNB phishing e-mail phishing was analyzed using the Google Admin Toolbox Message Header analyzer. The results showed SPF, DKIM, and DMARC

authentication failures, which indicate that the sender was not authorized to send emails on behalf of the domain.

The sender domain “fnb-securelogin.com” is not an official First National Bank domain and fails multiple email authentication checks. These failures strongly suggest phishing and email spoofing activity.

MessageId	002b224100c6228b@fnb-securelogin.com
Created at:	8/6/2025, 11:42:18 AM GMT+2 (Delivered after)
From:	First National Bank <support@fnb-securelogin.com>
To:	youremail@example.com
Subject:	Important Notice: Your Bank Account Has Been Locked
SPF:	fail with IP Unknown! Learn more
DKIM:	fail with domain fnb-securelogin.com; Learn more
DMARC:	fail Learn more

#	Delay	From *	To *	Protocol	Time received
0		unknown	→ bankmail.example.com	SMTP	8/6/2025, 11:42:18 AM GMT+2

PREVENTION & AWARENESS GUIDELINES

These are the best practices:

- Never click suspicious links
- Hover over the link before clicking
- Avoid downloading unknown attachments
- Report suspicious emails to IT/security teams immediately.

EMPLOYEES DO'S AND DON'TS

DO'S

- Verify unexpected emails
- Check domain spelling carefully
- Use strong passwords
- Report any suspicious messages

DON'TS

- Do not share OTPs or passwords
- Do not download random attachments
- Do not trust urgent scare tactics

Conclusion.

Phishing attacks remain one of the most dangerous forms of cybercrime because they target human behavior rather than technical vulnerabilities. Attackers use social engineering tactics such as urgency, urgency, and impersonation to manipulate victims into revealing sensitive information.

This project demonstrated how phishing emails can be analyzed by identifying suspicious domains, malicious links, urgent language, and authentication failures. By improving cyber security awareness and following safe online practices, users and organizations can reduce the risk of phishing attacks.

